Kuku Promo Finder Studio: Two-Stage Hybrid Ad Cut Engine

Problem 1: Find the Promos (Ads) | Dataset: *I Married a Broken Soldier. He was a Duke* (50 Episodes, 3,727 Cues, 102.8 Mins)

1. PROBLEM DECOMPOSITION & TWO-STAGE HYBRID ARCHITECTURE

Auditing all 50 screenplays (**100 scenes, 3,727 cues**) surfaced three structural traps that break naive single-prompt LLM pipelines: (1) **2,392 Multi-Track Overlaps** across concurrent `CAMERA`, `ACTION`, `SOUND FX` (114 cues), and `CHARACTER` (1,487 cues) timelines; (2) **22 Episode-End Cliffhangers** requiring lookahead to **Episode** `N+1` to verify the withheld payoff; and (3) **Crew Stage-Direction Lore**, where `CAMERA`/`ACTION` lines repeat character names that cold TikTok viewers never hear.

We decomposed the system into a **Two-Stage Hybrid Funnel** (`src/parser.py` → `src/candidates.py` → `src/llm_judge.py` → `src/benchmark.py`): **Stage 1 (`src/parser.py`, `src/candidates.py`)** separates cues into 4 concurrent tracks, normalizes scene-clock resets, and snaps cuts to **Dialogue-Safe Boundaries** (zero mid-line slices). It scores every valid `20.0s–60.0s` window across 5 sub-metrics (`0–20` each: `Hook Velocity`, `Conflict Escalation`, `Cliffhanger Peak`, `Standalone Clarity` via a **Spoken-Only Dynamic Speaker Lore Filter**, and `Pacing Fit`), deduplicates via **NMS (`IoU < 0.45`)**, and builds **Multi-Segment Stitches** (`multi_segment_ellipsis` & `cold_open`, each segment `>= 6.0s`, sum `20.0s–60.0s`). **Stage 2 (`src/llm_judge.py`)** evaluates top candidates in **5 macro-batches** (10 episodes/call) for status reversals and curiosity gaps, backed by a **SHA-256 disk cache** and **3-Layer Error Rescue** (JSON repair → backoff → Stage-1 fallback).

2. AI LEVERAGE, 1-FLAG MODEL SWAPPING & TOP PROMO CUTS

Deterministic Python enforces hard timestamp physics (`20.0s–60.0s`, dialogue-safe cuts, IoU deduplication), while Stage-2 AI evaluates narrative subtext. Models swap with zero code edits via CLI (`python cli.py --provider gemini|openai|anthropic|cli|semantic_reranker|heuristic`) or the Streamlit sidebar (`app.py`). In `raw_outputs/model_swap_diff.json`, Stage-2 re-ranking promotes cross-episode cliffhangers such as **Ep 43 (`+12` ranks to #1)** and **Ep 44 (`+13` ranks to #4)**. Sample picks from `raw_outputs/ranked_promos_top20.md` (`100%` within `20.0s–60.0s`):

- **#01 Ep 43 (`01:08.80–01:54.96`, 46.2s, Score `81.98`, `single`):** *Alexander: "You sold my men to the enemy." → Constable: "Going somewhere, Lord Sterling?"* **Payoff (`Ep 44 00:00–00:03.20`):** Constable blocks Sterling at the church archway.

- **#02 Ep 27 (`00:26.10–00:55.60`, 29.5s, Score `80.81`, `single`):** *Duchess: “My lost son wore that exact ring…” → Catherine: “What is that maid doing in Her Grace’s chamber?”* **Payoff (`Same Ep 00:55.60–01:02.00`):** Juliana hides her hands under suspicion.
- **#03 Ep 04 (`01:50.00–02:29.00`, 39.0s, Score `80.63`, `single`):** *Priest: “I now pronounce you…” → Bearded Man: “Come, Lady Juliana.”* + oak doors creak open. **Payoff (`Ep 05 00:00–00:02.50`):** Juliana walks out clutching the heirloom ledger.
- **#06 Ep 03 (`[00:00.00–00:16.30] + [01:49.00–02:20.30]`, 47.6s, Score `80.28`, `ellipsis_stitch`):** Bridges altar betrayal setup with final marriage vow climax, cutting 92.7s of middle exposition.

3. QUALITY JUDGEMENT: WHERE NAIVE AI GOT IT WRONG (`RAW_OUTPUTS/QUALITY_AUDIT_FAILURES.MD`)

- **(1) Concurrent Track Guillotine (`Ep 01`):** Naive AI picked `00:11.00–00:52.40`, slicing mid-`CAMERA` shot (`00:52.00–00:54.60`). *Fix:* 4-track parser + `is_dialogue_safe_timestamp()` snaps to `00:54.60–01:38.20` (**54% → 100% dialogue-safe cuts**).
- **(2) Stage-Direction Lore False Positive (`Ep 14, 22`):** Penalized action clips because crew `CAMERA` notes repeated character names. *Fix:* Dynamic speaker lore filter (`17` names) applied **strictly to spoken** `DIALOGUE` (**65% → 95% cold-viewer clarity**).
- **(3) 0-Second Scene-End Payoff Blindspot (`Ep 36, 37, 43`):** Left payoff blank at `scene_end` (`0.0s` left). *Fix:* Cross-episode lookahead pulls `Episode N+1` opening cues + `CONTINUITY NOTES` (**48% → 100% verified payoffs**).
- **(4) Corridor Transition Dead Air (`Ep 03, 18, 30`):** Contiguous window included 25s walking lulls. *Fix:* `multi_segment_ellipsis` & `cold_open` stitches + 1-click FFmpeg `filter_complex` concat script (`raw_outputs/ffmpeg_slice_commands.sh`).

4. PROTOTYPE GUIDE & 100X SCALE THINKING (1 SHOW TO 200 SHOWS/WEEK)

Run `python cli.py` (all 50 episodes in **6.5s**) or `streamlit run app.py` (4-tab Studio with `.zip`/`.txt` upload, custom duration sliders for shortlist interview changes, Model-Swap Diff, and 1-click CSV/FFmpeg exports; verified by `15/15` `pytest` tests). Manual promo cutting takes ~15 hrs/show (`$450/show`); unbatched LLM scanning costs `$4.20/show`. Our hybrid funnel costs **`$0.015/show` (5 macro-batched calls)**, saving **99.6% cost** and **1,450 editor hours/week** at 100 shows/week. At 100x scale, we add **Whisper word-level prosody + PySceneDetect shot boundaries** and feed back **TikTok/Meta 3s Thumbstop & IPM** into Stage-2 calibration.